

Question 1 What is Object-Oriented Programming (OOP)?

Answer: Object-Oriented Programming (OOP) is a programming paradigm based on the concept of objects. It uses classes and objects to organize code and allows features such as inheritance, encapsulation, polymorphism, and abstraction.

Question 2 What is the primary purpose of a database management system (DBMS)?

Answer: A Database Management System (DBMS) is software used to create, manage, and manipulate databases. Its primary purpose is to efficiently store, retrieve, update, and secure data while reducing redundancy and improving consistency.

Question 3 Explain the difference between HTTP and HTTPS.

Answer: HTTP (HyperText Transfer Protocol) is used for communication between web browsers and servers, but it does not encrypt data.

Question 4 What is Artificial Intelligence?

Answer: Artificial Intelligence (AI) is the simulation of human intelligence in machines. AI systems can learn from data, solve problems, recognize patterns, and make decisions with minimal human intervention.

Question 5 What is the role of an operating system?

Answer: An operating system manages computer hardware and software resources. It provides an interface between the user and hardware, manages memory, processes, files, and controls input/output devices.